such as ensuring that uniform road signs with speed limits appear at regular intervals with night glow paints to installing crash barriers and lane markings with road studs. The interim measures focused on extra-widening at curves in blackspot locations, signalisation and removal of obstructions. The long-term measures include providing bypasses, service roads and grade-separated inner-changes with pedestrian underpasses and foot overbridges. Meanwhile, it was made mandatory that all new highways get a road safety clearance at the design stage itself before getting approval for the detailed project report.

India is also unique in other respects. Buses, for instance, are involved in 5-10 per cent of the total fatal road crashes compared to 0.5-1.6 per cent in developed nations. Also, unlike in most countries, Indian highways intersect urban clusters. The mix of local commuters, pedestrians and high-speed moving traffic increases the likelihood of crashes at locations where service lanes have been encroached upon or which do not have enough underpasses or overbridges. In such zones, Gadkari pushed for traffic-calming measures such as humps on the approach road, warning signs, delineators and bar markings. Rising pedestrian fatalities on highways were addressed by allowing regional NHAI officers to use up to Rs 25 crore to construct pedestrian underpasses and subways and Rs 1.25 crore for foot overbridges.



"In most cases, [accident victims] are dumped into an ambulance and taken to a hospital with little care given in transit.... Ambulance care isn't just about transfer but transit care as well"

PIYUSH TEWARI
Founder, SaveLIFE Foundation

TOUGHENING VEHICLE DESIGN NORMS

Lack of proper fitness certifications and monitoring of vehicles, especially trucks, buses and cars, is another significant cause of accidents on Indian highways. Thus tyre bursts, brake failures and loss of steering control end up in crashes. Since transport is on the concurrent list, state government transport officials issue fitness certificates. However, once issued for new vehicles, there are hardly any follow-ups for vehicle maintenance, barring pollution checks. Manufacturers, too, are selective about equipping vehicles with higher safety features such as airbags or designing vehicle bodies to lessen the impact of a crash, including using fire-resistant material.

Gadkari has in the past few years initiated several steps to ensure improvements in vehicle safety. In 2021, his ministry announced that all commercial vehicles, like trucks, buses and taxis, have to be scrapped once they are 15 years old and private vehicles once they turn 20. The Vehicle Scrapping Policy was to get old and unfit vehicles off the roads, especially on highways, as well as reduce vehicular pollution by getting manufacturers to switch to engines using less polluting fuel.



MILIND SHELTE

CASE STUDY

ROHAN MISHRA, 28, *Mumbai*

WHAT HAPPENED

Our family's 2013 Mumbai–Delhi road trip was disrupted by a crash near Jaipur at dusk. A truck carelessly merged into our lane, forcing my father to brake abruptly. Our vehicle's driver's side slammed into the truck. Despite no physical injuries, we endured a frightening few hours on the poorly lit highway. My father's friends from Jaipur eventually arrived, towing our vehicle to a repair shop.

HOW HIGHWAYS CAN BE MADE SAFE

As a regular long-distance driver, I've noticed persistent issues: drivers neglecting indicators and stop signs, plus rampant misuse of high beams. All this makes night driving especially treacherous. Addressing these would greatly enhance road safety.